🐍 PYTHON LEARNING ROADMAP JOURNEY

Name: Christabel
Goal: Become a skilled programmer and Computer Science student
Main Programming Language: Python
Started Learning: 2026

🌟 MY PYTHON LEARNING GOAL

My goal is to learn Python from the basics to an advanced level. I want to understand how programming works, build my own projects, and eventually use my skills to create websites, games, applications, and other useful software.

I will learn step by step, practice regularly, build projects, review what I have learned, and combine old knowledge with new knowledge instead of simply moving from topic to topic.

🗺️ MY PYTHON ROADMAP

LEVEL 1 — PYTHON BASICS ✅

Topics

- What Python is
- Installing Python
- Using VS Code
- Creating and running Python files
- `print()`
- Variables
- `input()`
- Strings
- Integers
- Floats
- `int()`
- `float()`
- Basic arithmetic
- Comments
- Comparison operators

Practice Projects

- Greeting program
- Name and hobby program

- Simple calculator

Status: ✅ Core topics learned

LEVEL 2 — DECISION MAKING ✅

Topics

- Boolean values
- `True` and `False`
- `if`
- `elif`
- `else`
- Logical operators:
- `and`
- `or`
- `not`
- Nested `if` statements

Practice Projects

- Grade checker
- Age checker
- Simple quiz
- Number guessing game

Status: ✅ Learned and practiced

LEVEL 3 — LOOPS ✅

Topics

- `for` loops
- `while` loops
- `range()`
- `break`
- `continue`
- `while True`
- Counting with loops
- Looping through collections
- Combining loops with conditions

Practice Projects

- Multiplication table

- Number counter
- Guessing game
- Quiz game

Status: ✅ Learned and practiced

LEVEL 4 — LISTS AND LIST OPERATIONS ✅

Topics

- Creating lists
- Accessing list items
- Changing list items
- `.append()`
- `.remove()`
- `.pop()`
- `len()`
- `enumerate()`
- Looping through lists

Practice Projects

- Name list program
- Shopping list
- Task manager
- Simple contact list

Status: ✅ Core list concepts learned

Note: Some list-related built-in functions and methods may be reviewed again later.

LEVEL 5 — FUNCTIONS ✅

Topics Learned

- What functions are
- Creating functions with `def`
- Calling functions
- Parameters
- Arguments
- Parameters vs arguments
- Positional arguments
- Keyword arguments
- Positional vs keyword arguments
- Default parameters

- `return`
- Local variables
- Global variables
- `*args`
- `**kwargs`
- Using functions with `input()`
- f-strings
- Combining functions with loops, lists, conditions, and input

Important Terminology

Parameter: A placeholder in a function definition.

Argument: The actual value passed to a function.

Positional argument: An argument matched according to its position.

Keyword argument: An argument matched using the parameter name.

***args:** Collects extra positional arguments into a tuple.

****kwargs**:** Collects extra keyword arguments into a dictionary.

Status: ✅ Core Functions topic complete

LEVEL 6 — DICTIONARIES AND TUPLES 🔄

Dictionaries

Topics to learn:

- What dictionaries are
- Keys and values
- Creating dictionaries
- Accessing dictionary data
- Adding dictionary items
- Changing dictionary items
- Removing dictionary items
- `keys()`
- `values()`
- `items()`
- Looping through dictionaries
- Using dictionaries with functions

Tuples

Topics to learn:

- Creating tuples
- Accessing tuple items
- Tuple immutability
- When to use tuples
- Difference between lists and tuples

Practice Projects

- Student information system
- Contact book
- Simple inventory program

Status: 🔄 Next major topic

LEVEL 7 — SETS AND MORE DATA STRUCTURES

Topics

- Sets
- Set operations
- Understanding different data structures
- Choosing the right data structure for a problem

Practice Projects

- Duplicate remover
- Simple data organizer

Status: ⬜ Not started

LEVEL 8 — FILE HANDLING

Topics

- Opening files
- Reading files
- Writing files
- Appending to files
- Closing files
- Working with `.txt` files
- Introduction to JSON

Practice Projects

- Notes app
- To-do list that saves tasks
- Simple student record system

Status: ⬜ Not started

LEVEL 9 — ERROR HANDLING

Topics

- Common Python errors
- `try`
- `except`
- `else`
- `finally`
- Debugging programs
- Input validation

Practice Projects

- Error-resistant calculator
- Input validation program

Status: ⬜ Not started

LEVEL 10 — OBJECT-ORIENTED PROGRAMMING (OOP)

Topics

- Classes
- Objects
- Attributes
- Methods
- `__init__`
- Inheritance
- Encapsulation

Practice Projects

- Student management system
- Bank account simulation
- Simple game using classes

Status: ⬜ Not started

LEVEL 11 — ADVANCED PYTHON 🚀

Topics to Explore

- Modules
- Packages
- Importing code
- Virtual environments
- Lambda functions
- List comprehensions
- Iterators
- Generators
- Decorators
- Python Standard Library

Useful modules to encounter later

- `os`
- `math`
- `random`
- `datetime`
- Other useful standard-library modules

Status: ⬜ Not started

🌐 LEVEL 12 — PYTHON FOR WEB DEVELOPMENT

Topics to Learn

- Flask
- Django
- APIs
- Databases
- SQL
- Connecting Python to websites
- Backend development

Projects

- Personal portfolio website
- Website for Dad's business
- Blog website
- Simple web application

Status: ⬜ Future topic

🎮 LEVEL 13 — PYTHON FOR GAME DEVELOPMENT

Topics to Explore

- Game logic
- Game loops
- Pygame
- Game design basics
- Graphics
- Sound
- Player controls

Project Goal

Create my own original game.

Status: 🔄 Exploring game development

🧠 LEVEL 14 — COMPUTER SCIENCE AND PROGRAMMING

Topics to Learn

- Algorithms
- Data structures
- Problem solving
- Computational thinking
- Git and GitHub
- Basic computer science concepts
- Competitive programming skills

Competition Preparation

As I become stronger, I want to practice:

- Logic problems
- Pattern problems
- Searching and sorting
- Recursion
- Time and space complexity
- Algorithms
- Data structures
- Problem-solving under pressure

Goal: Become ready to participate in programming and technology competitions during secondary school.

💻 MY FUTURE PROGRAMMING LANGUAGES AND TECHNOLOGIES

After becoming comfortable with Python, I plan to explore:

1. HTML
2. CSS
3. JavaScript
4. C
5. C++
6. C#
7. Unity

Other technologies I may encounter later

- Java
- SQL
- Kotlin
- Swift
- Dart
- TypeScript
- R
- PHP
- React
- Node.js
- Django
- Flask
- PyTorch
- TensorFlow
- AWS
- Azure
- Google Cloud
- Docker
- Kubernetes
- Linux
- Power BI
- Tableau
- Blender

Important: I do not need to learn all of these at once. They belong to different career paths and will be introduced when they become useful.

🏆 MY PROJECT JOURNEY

Beginner Projects

☐ Greeting Program
☐ Calculator
☐ Grade Checker
☐ Number Guessing Game
☐ Multiplication Table

Intermediate Projects

☐ To-Do List
☐ Quiz Game
☐ Contact Book
☐ Notes App
☐ Student Management System

Bigger Projects

☐ Personal Portfolio Website
☐ Website for Dad's Business
☐ Original Game
☐ Useful Python Application

Completed / Built During Learning

☑️ Career Profile Program

📚 MY LEARNING METHOD

I learn best when I:

- Learn a concept first
- See a worked example
- Understand what the code is doing
- Try my own practice task
- Receive hints when I am stuck
- Combine the new concept with older knowledge
- Build projects that use several topics together
- Review concepts that were introduced but not properly taught

A major lesson from my learning so far is:

Do not confuse "used once" with "fully learned."

Some methods, built-ins, and modules may appear naturally in projects before I study them properly. I will keep track of those and return to them under their correct topics.

🗂️ MY NOTES AND GITHUB JOURNEY

I am saving my learning notes and programming work in Git and GitHub.

My repository can contain:

- Notes
- Practice code
- Projects
- Challenges and solutions
- Roadmaps
- Progress trackers
- Project ideas
- README files

This means my GitHub can gradually become a record of my programming growth and an early portfolio.

🌟 MY BIG DREAM

I want to become an excellent programmer and study Computer Science at university.

I want to use my programming skills to:

- Create amazing projects
- Build useful software
- Participate in coding competitions
- Earn medals, recognition, and achievements
- Build a strong portfolio
- Explore game development
- Explore AI and data
- Eventually work in technology
- Possibly study Computer Science at UNILAG
- Consider strong university opportunities in Canada or the United States if they become available

I also want to make my parents proud through my effort, consistency, growth, and achievements.

I know that becoming a great programmer takes time, patience, and practice.

I will keep learning, keep building, keep reviewing, and keep improving.

This is the beginning of my Python journey. 🐍💻🚀

Learn. Build. Solve. Compete. Grow.